

FPGA-Based Digital Down-Conversion (DDC) System

Design and Implementation Requirements Specification

1 Project Background

This project implements a digital down-conversion (DDC) system using an FPGA, including DDS/NCO, digital mixing, FIR low-pass filtering, decimation, and spectrum analysis to convert an intermediate-frequency signal to a baseband signal.

2 Overall System Architecture

ADC Data Source → DDS/NCO → Digital Mixer → FIR Lowpass Filter → Decimator → Baseband IQ

3 Functional Requirements

FR-1 DDS Local Oscillator Generation: $F_s=100\text{MHz}$, DDS frequency 20MHz, phase width 32bit, output width 12bit.

FR-2 Digital Mixing: $I=IF*\cos$, $Q=IF*\sin$.

FR-3 FIR Lowpass Filter: 32 Tap, passband 1MHz, stopband 5MHz, stopband attenuation 40dB.

FR-4 Decimation: Reduce 100MHz to 10MHz, decimation factor 10.

FR-5 FFT Analysis: Implement spectrum analysis in Python.

4 Input/Output Specifications

Input: 20MHz intermediate-frequency signal + 35MHz interference signal.

Output: I/Q data, sampling rate 10MHz.

5 Performance Specifications

Frequency error <100Hz; amplitude error <5%; SNR>35dB.

FPGA resources: LUT<20%, FF<15%, DSP<20.

6 Python Model Requirements

DDS model, Mixer model, FIR model, FFT analysis model.

7 RTL Design Requirements

Modules: dds.v, mixer.v, fir_filter.v, decimator.v, ddc_top.v.

8 Testbench Requirements

DDS verification, FIR verification, DDC system verification, comparison of RTL and Python Golden results.

9 FPGA Verification Requirements

Use ILA to capture data at key nodes; export IQ data via UART for spectrum analysis.

10 Graduation Project Deliverables

Proposal, midterm report, graduation thesis, defense PPT, Python code, RTL code, Testbench, and board-level verification results.